

Diagnostic Performance of an Integrated 3D Vascular Reconstruction and Aneurysm Detection AI Model using MR angiography

Purpose: To develop and validate a deep learning-based framework for automated cerebral aneurysm detection on MR angiography (MRA) and compare its diagnostic performance with experienced neuroradiologists using an external validation cohort.

Materials and Methods: A total of 3,036 MR-TOF datasets containing 4,360 annotated aneurysms were used to train the AI framework. The system consisted of MR-Vessel AI for high-resolution vascular reconstruction and MR-Aneurysm AI for lesion detection using 3D sub-volume analysis and iterative refinement. External validation was performed on 100 cases with 150 DSA-confirmed aneurysms across 19 anatomical locations and compared with interpretations by three neuroradiologists.

Results: The AI model achieved significantly higher sensitivity than the radiologist group (92.0% vs. 68.7%, $p = 0.0324$), while maintaining high specificity (97.8%). Negative predictive value was also superior in the AI model (99.3% vs. 97.4%). In subgroup analysis, the AI demonstrated excellent detection performance in the paraclinoid artery (96.0% vs. 65.3%) and anterior communicating artery aneurysms (87.5% vs. 79.2%, $p = 0.0062$).

Conclusion: The proposed deep learning framework showed robust performance for automated cerebral aneurysm detection on MRA, particularly with high sensitivity and NPV. The system may serve as an effective clinical decision-support tool for neurovascular screening by reducing diagnostic oversight and inter-observer variability.